

RCIM Model-Bank Reproduction Backward Paper-Faithful Grid-Search Closeout

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: track1_bidirectional_paper_faithful_grid_search_campaign_2026-05-04_12_26_30
- started at: 2026-05-14T00:44:05+02:00
- finished at: 2026-05-16T19:04:25+02:00
- closed direction: backward
- refreshed archive root: models/paper_reference/rcim_track1/backward
- benchmark report: doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md

Family Results

Family	Run Instance	Mean MAE	Mean RMSE	Mean MAPE %	Exported ONNX	Exported PKL
SVM	2026-05-15-23-28-01_track1_paper_faithful_grid_search_backward_svr_campaign_validation	0.22957	0.368131	48.8009	19	19
MLP	2026-05-15-23-30-56_track1_paper_faithful_grid_search_backward_mlp_campaign_validation	0.245743	0.349376	3411.06	19	19
RF	2026-05-15-23-47-53_track1_paper_faithful_grid_search_backward_rf_campaign_validation	0.0642822	0.144738	23.0516	19	19
DT	2026-05-16-01-10-50_track1_paper_faithful_grid_search_backward_dt_campaign_validation	0.0432093	0.134447	24.1088	19	19
ET	2026-05-16-01-12-39_track1_paper_faithful_grid_search_backward_et_campaign_validation	0.0858495	0.216616	36.2701	19	19
ERT	2026-05-16-01-14-27_track1_paper_faithful_grid_search_backward_ert_campaign_validation	0.0496713	0.129551	20.12	19	19
GBM	2026-05-16-03-42-16_track1_paper_faithful_grid_search_backward_gbm_campaign_validation	0.0603563	0.137289	23.0803	19	19
HGBM	2026-05-16-05-10-28_track1_paper_faithful_grid_search_backward_hgbm_campaign_validation	0.0876331	0.170341	32.8818	19	19
XGBM	2026-05-16-07-29-43_track1_paper_faithful_grid_search_backward_xgbm_campaign_validation	0.113797	0.191087	43.5059	19	19
LGBM	2026-05-16-07-33-09_track1_paper_faithful_grid_search_backward_lgbm_campaign_validation	0.067018	0.13972	24.0255	19	19
ELM	2026-05-16-19-03-18_track1_paper_faithful_grid_search_backward_elm_campaign_validation	0.237422	0.337425	71.0273	19	19

Benchmark Status

Table	Green	Yellow	Red	Total
Backward Table 2 - Amplitude MAE	61	22	27	110
Backward Table 3 - Amplitude RMSE	63	20	27	110
Backward Table 4 - Phase MAE	65	21	13	99
Backward Table 5 - Phase RMSE	65	21	13	99

Reference Archive Refresh

Family	Archived Targets	Source Runs	Archive Root
SVM	19	1	models/paper_reference/rcim_track1/backward/svm_reference_models
MLP	19	1	models/paper_reference/rcim_track1/backward/mlp_reference_models
RF	19	1	models/paper_reference/rcim_track1/backward/rf_reference_models
DT	19	1	models/paper_reference/rcim_track1/backward/dt_reference_models
ET	19	1	models/paper_reference/rcim_track1/backward/et_reference_models
ERT	19	1	models/paper_reference/rcim_track1/backward/ert_reference_models
GBM	19	1	models/paper_reference/rcim_track1/backward/gbm_reference_models
HGBM	19	1	models/paper_reference/rcim_track1/backward/hgbm_reference_models
XGBM	19	1	models/paper_reference/rcim_track1/backward/xgbm_reference_models
LGBM	19	1	models/paper_reference/rcim_track1/backward/lgbm_reference_models
ELM	19	1	models/paper_reference/rcim_track1/backward/elm_reference_models

Best Campaign Representative

- run: track1_paper_faithful_grid_search_backward_dt
- family: DT
- mean MAE: 0.0432093
- mean RMSE: 0.134447

Notes

- The original paper-family order for Tables 2 - 5 remains unchanged.
- ELM is archived and benchmarked as an operational RCIM Model-Bank Reproduction family because the completed campaign includes it.
- Forward RCIM Model-Bank Reproduction paper-reference archives were not modified by this backward-only closeout.